

LAST NAME: _____

Practice Quiz 07D — MATH 2055 — Fall 2026

Practice for Quiz 07

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

1. Construct a function $f : \mathbf{R} \rightarrow \mathbf{R}$ such that

$$f(0) < 0 \quad \text{and} \quad \lim_{x \rightarrow 0} f(x) = 1.$$

Give a formula defining f for every real input, and prove the limit directly from the epsilon-delta definition.